

Info on any aircraft

- aircraft speed
- angle of attack
- Reynolds number $\rightarrow R_e = \frac{\rho V L}{\mu}$, μ = dynamic viscosity. How smooth / chaotic the air plane is.

- Wing reference area
- mean aerodynamic chord
- reference wing span

- Center of gravity location
- Vertical COG
- Wing longitudinal position
- Wing vertical position
- Wing incidence angle

- Number of body stations
- Longitudinal station positions
- body radii
- body cross sectional areas
- body perimeters
- nose/body length
- nose type
- aft body length

- Root chord
- Tip chord
- Semispan
- effective semispan
- wing sweep
- Chord reference location
- Flap sweep
- wing twist
- break semispan
- Inner dihedral

Flight Condition Input

Reference Geometry

$$mac = \frac{2}{3} C_r \frac{1 + \lambda + \lambda^2}{1 + \lambda}, \quad C_r = \text{root chord}, \quad \lambda = \frac{C_t}{C_r}$$

Wing chord for stability CG calc.

Aircraft layout / CG

- Angle between wing chord line & aircraft fuselage reference line.
- Station are measurements along fuselage length. Use to define aircraft geometry.

Fuselage Geometry

- Partition of each station
- Area of fuselage slice as a station, $A = \pi R^2$
- Circumference around fuselage section $P = 2\pi R$
- length of front fuselage section
- Contour, oval, round, or parabolic

Wing Geometry

- The chord near the fuselage

Outer Dihedral

- wing type
- Thickness-to-chord ratio
- max thickness position
- design lift coefficient
- design alpha
- lift curve slope
- maximum lift coefficient
- zero-lift pitching moment
- leading edge radius
- CAMBER: whether wing is cambered
- Lift at zero alpha
- Trailing-edge flap effectiveness

Airfoil / wing Section Aerodynamics

Simulation with Dascam

- CASEID: simulation label
- Save: store case results
- Dump case: output detailed data
- Next case: Move to next simulation case

Tail Position Params

- XA: Horizontal tail x-position
- ZH: vertical position
- ALH: Horizontal tail incidence angle
- HTPZNF: Horizontal tail planform definition
- CHRDR: Root chord of the tail
- CHRTP: Tip chord of the tail
- SSPW: Half the tail span
- SSPNE: Exposed semi span

Geometry